

Fused-head determinatives

Fragment 1 — 2026-05-07

This fragment tests the local end of the project: can a CGEL-style category / function distinction be stated as a small constraint on labelled trees? The target is the familiar contrast between an NP with a separate nominal head and an NP in which a determinative supplies the head-like material itself.

1. Phenomenon

CGEL treats examples such as *three*, *some*, and *many* in independent NP use as cases where a determinative is not followed by an overt nominal head.

- (1)
- (2) *Pat invited two students; I invited three.*
- (3) *Some of the reports were late, but many arrived on time.*
- (4) *These are useful; those are defective.*

The descriptive problem is small but useful. In *two students*, the determinative fills the Det function and the nominal fills Head. In *three*, a single determinative item projects the substantive NP by itself. A fragment should therefore distinguish ordinary Det-plus-Head NPs from determinative-headed NPs without collapsing category and function.

2. Vocabulary

Use the shared relations $\triangleleft$ and $\triangleleft^*$, and the function relations Head and Det. Add one local lexical-category predicate:

- (5) $\text{Dtr}(x)$: x is a determinative-category node.

For this draft, $\text{Fused}(d, n)$ is a bookkeeping predicate saying that the determinative node d inherits whatever interpretation would ordinarily be supplied by the missing nominal material in NP n . The point is not yet to formalise recoverability; it is to isolate the tree configuration that calls for it.

3. Specification

An ordinary NP with a separate nominal head can be characterised as follows:

- (6) $\text{CanonicalNomNP}(n) \equiv_{\text{def}} \text{NP}(n) \wedge (\exists h)[n \triangleleft h \wedge \text{Head}(n, h) \wedge \text{Nom}(h)]$.
an NP whose Head daughter is nominal

The fused-head determinative configuration is the corresponding case where the Head daughter is determinative and no separate nominal Head daughter is present:

- (7) $\text{DHeadNP}(n) \equiv_{\text{def}} \text{NP}(n) \wedge (\exists d)[n \triangleleft d \wedge \text{Head}(n, d) \wedge \text{Dtr}(d) \wedge \text{Fused}(d, n)]$
 $\wedge \neg(\exists h)[n \triangleleft h \wedge \text{Head}(n, h) \wedge \text{Nom}(h)]$.
an NP headed by a determinative rather than by an overt nominal

For simple cases such as *three*, this is a first-order local definition. Partitive cases add one more daughter:

- (8) $\text{PartitiveDHeadNP}(n) \equiv_{\text{def}} \text{DHeadNP}(n) \wedge (\exists p)[n \triangleleft p \wedge \text{Comp}(n, p) \wedge \text{PP}(p)]$.
a determinative-headed NP with an of-PP complement

Nothing in the formula says that determinatives are nouns, pronouns, or a separate primary category. It says only that this tree has a determinative node in Head function.

4. Worked instance

Take *I invited three*. Let n be the object NP and d the lexical node *three*. In the intended tree, $n \triangleleft d$, $\text{Head}(n, d)$, and $\text{Dtr}(d)$. There is no daughter h such that $\text{Head}(n, h)$ and $\text{Nom}(h)$. Therefore n satisfies $\text{DHeadNP}(n)$.

The contrast with *two students* is immediate. In that NP, the determinative *two* fills $\text{Det}(n, d)$, while the nominal *students* fills $\text{Head}(n, h)$. That NP satisfies $\text{CanonicalNomNP}(n)$, not $\text{DHeadNP}(n)$.

5. Open questions

- (a) This draft treats Fused as primitive. A fuller fragment needs a recoverability condition for the missing nominal contribution, or a decision that the simple cases should be analysed as ordinary determinative heads without fusion.
- (b) Partitives probably require tighter treatment of the *of*-PP: it is not just any complement PP.
- (c) The defective articles *the* and *a* should be kept outside the productive simple-head generalisation unless their special status is explicitly modelled.
- (d) CGEL page references and the Payne–Huddleston–Pullum treatment of fused functions need to be checked before this becomes a citable fragment.